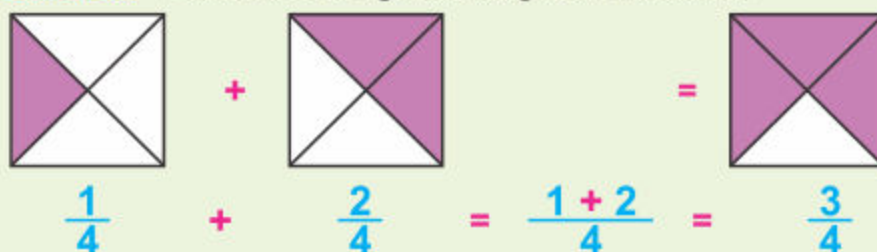


Represent addition of fractions through figures

Let us perform addition of two fractions by their coloured parts.

Example 1: Add $\frac{1}{4}$ and $\frac{2}{4}$.

Solution: Look at the figures of given fractions

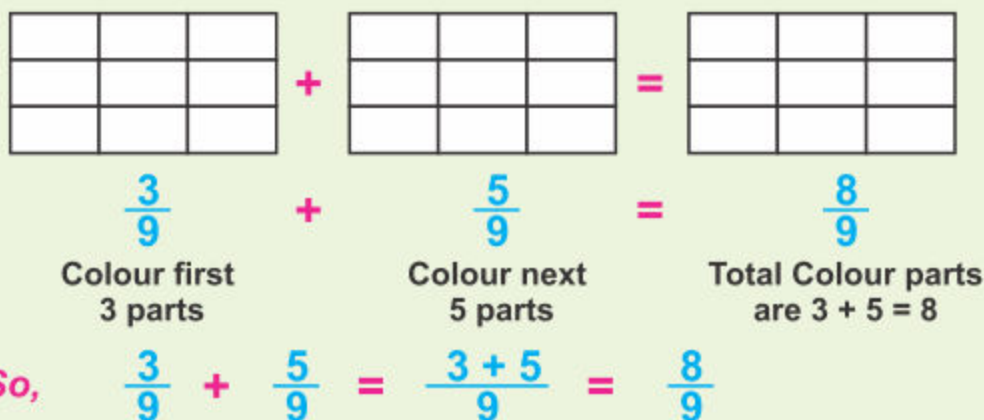


i.e one-fourth and two-fourth give three fourth.



Activity

Add $\frac{3}{9}$ and $\frac{5}{9}$ by colouring the shapes of fractions.

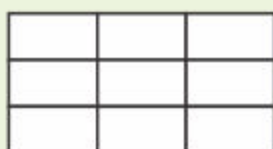


Teacher's Note

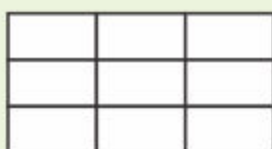
Teacher should help the students to add the fraction having same denominators.

**Activity**

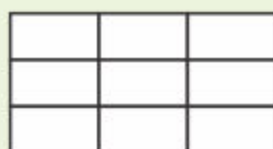
Add $\frac{3}{9}$ and $\frac{5}{9}$ by colouring the shapes of fractions.



+



=



$$\frac{3}{9}$$

+

$$\frac{5}{9}$$

=

$$\frac{8}{9}$$

Colour first
3 parts

Colour next
5 parts

Total Colour parts
are $3 + 5 = 8$

So, $\frac{3}{9} + \frac{5}{9} = \frac{3+5}{9} = \frac{8}{9}$

EXERCISE 29

(A) Solve the following fractions:

(1) $\frac{1}{3} + \frac{1}{3}$

(2) $\frac{1}{5} + \frac{3}{5}$

(3) $\frac{2}{4} + \frac{1}{4}$

(4) $\frac{3}{6} + \frac{2}{6}$

(5) $\frac{2}{7} + \frac{4}{7}$

(6) $\frac{3}{8} + \frac{4}{8}$

(7) $\frac{5}{9} + \frac{4}{9}$

(8) $\frac{3}{11} + \frac{6}{11}$

(9) $\frac{3}{15} + \frac{10}{15}$

(10) $\frac{11}{18} + \frac{6}{18}$

(11) $\frac{1}{14} + \frac{12}{14}$

(12) $\frac{9}{20} + \frac{10}{20}$

(B) Add:

(1) $\frac{3}{7}$ and $\frac{2}{7}$

(2) $\frac{4}{12}$ and $\frac{5}{12}$

(3) $\frac{6}{15}$ and $\frac{7}{15}$

(C) Add the following fractions through figures:

(1) $\frac{2}{5} + \frac{1}{5}$

(2) $\frac{3}{7} + \frac{5}{7}$

(3) $\frac{8}{10} + \frac{1}{10}$

(4) $\frac{1}{12} + \frac{1}{12}$

(5) $\frac{6}{13} + \frac{3}{13}$

(6) $\frac{7}{15} + \frac{4}{15}$

3.6 SUBTRACTION OF FRACTIONS

Subtract fractions with same denominators

Starter: The teacher will write the given example on the board and explain how to subtract fractions when the denominator is the same. The teacher will call different children to the board and have them solve the sums given in the Activity.

To subtract the fractions having same denominators, we will subtract their numerators only.

Hence

Subtraction of two fractions having same denominators



Subtraction of numerators
Denominator

Example 1: Subtract $\frac{3}{4}$ from $\frac{3}{4}$

Solution: $\frac{3}{4} - \frac{3}{4} = \frac{3-3}{4} = \frac{0}{4} = 0$

Activity

Solve.

(1) $\frac{3}{4} - \frac{2}{4} = \frac{3-2}{4} = \square$

(2) $\frac{6}{7} - \frac{1}{7} = \frac{6-1}{7} = \square$

(3) $\frac{11}{12} - \frac{5}{12} = \frac{11-5}{12} = \square$

Represent subtraction of fractions through figures

Let us perform subtraction of two fractions by colouring the respective parts.

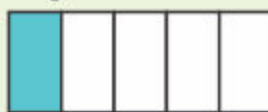
Example 1: Subtract $\frac{2}{5}$ from $\frac{3}{5}$.

Solution: Consider the figures of given fractions.



$$\frac{3-2}{5}$$

=



$$\frac{1}{5}$$

i.e. From $\frac{3}{5}$ subtracted $\frac{2}{5}$ and get $\frac{1}{5}$.

EXERCISE 30

1. Subtract the following fractions:

(i) $\frac{2}{3} - \frac{1}{3}$

(ii) $\frac{4}{5} - \frac{2}{5}$

(iii) $\frac{5}{6} - \frac{4}{6}$

(iv) $\frac{6}{8} - \frac{3}{8}$

(v) $\frac{5}{7} - \frac{3}{7}$

(vi) $\frac{7}{9} - \frac{5}{9}$

(vii) $\frac{8}{10} - \frac{3}{10}$

(viii) $\frac{9}{13} - \frac{5}{13}$

(ix) $\frac{6}{11} - \frac{1}{11}$

(x) $\frac{10}{12} - \frac{3}{12}$

(xi) $\frac{13}{15} - \frac{12}{15}$

(xii) $\frac{15}{18} - \frac{10}{18}$

2. Subtract:

(i) $\frac{1}{4}$ from $\frac{2}{4}$

(ii) $\frac{2}{14}$ from $\frac{13}{14}$

(iii) $\frac{16}{19}$ from $\frac{17}{19}$

3. Subtract the following fractions and draw the figures:

(i) $\frac{5}{7} - \frac{3}{7}$

(ii) $\frac{8}{9} - \frac{1}{9}$

(iii) $\frac{4}{10} - \frac{3}{10}$

(iv) $\frac{12}{13} - \frac{5}{13}$

(v) $\frac{9}{11} - \frac{5}{11}$

(vi) $\frac{10}{17} - \frac{4}{17}$

(vii) $\frac{13}{25} - \frac{7}{25}$

(viii) $\frac{16}{17} - \frac{15}{17}$

(ix) $\frac{11}{15} - \frac{8}{15}$

4.1 LENGTH

We have learnt about length and the unit of length i.e metre and centimetre.

Starter: The teacher will do a quick recap by writing the units of length and their symbols on the board and have children identify them.

Read standard units of length (kilometre, metre and centimetre) including abbreviations.

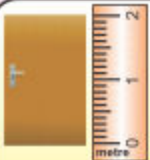
The units of length are used to measure height, width, depth, length and distance. In the international system, of standard unit, the metre is the basic unit of length. It is denoted by m.

We use centimetres for measuring small lengths, metres for bigger and kilometres for long distance. The abbreviation used for kilometre is "km", for metre is "m" and for centimetre is "cm".



Activity

Colour the correct unit of length.

 Ant  cm m km	 Height of Tree cm m km	 Height of Mountain cm m km
 Width of Phuleli Canal cm m km	 Length of Pen cm m km	 Height of Door cm m km
  Distance from Karachi to Sukkur cm m km		

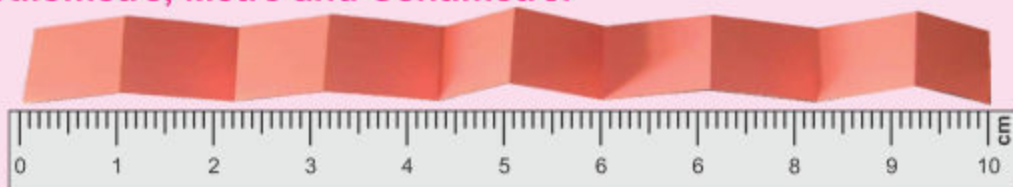
Teacher's Note

Teacher should revise the standard units of length and their abbreviations.

Measure and write standard units of length including abbreviations.

The teacher will ask students to make these strips and then have them work in pairs and then groups to measure the lengths of various objects in the classroom.

Kilometre, Metre and Centimetre.



The length of this strip is 10 centimetres.

Make 10 of these strips. Tape them together.

The length of your new strip is one **metre (m)**.

$$100 \text{ centimetres} = 1 \text{ metre}$$

Now use your metre strip to measure the length and height of your desk and height of a door.

Suppose you made 1000 meter strips and taped them together.

The length of your new strip would be one kilometre (km).

$$1000 \text{ metres} = 1 \text{ kilometre}$$

Note: Kilometre is used to measure the long distance.



Sukkur

The distance between Sukkur and
Karachi city is 490 km.



Karachi

Teacher's Note

Teacher should help the students to measure the exact unit of length.

EXERCISE 31

(1) Measure and write the length of the following:

(i)  6 cm

(ii)  _____

(iii)  _____

(iv)  _____

(v)  _____

(2) Tick (✓) the correct and Cross the (✗) the in correct answer:

(i) Length of the  is measured in metres. ☒

(ii) Length of the  is measured in metres. ☐

(iii) Distance from



Karachi

to



Hyderabad

is measured in **km**.

(iv) Height of the



is measured in **centimetres**.

(v) Length of the



is measured in **kilometres**.

(vi) Length of the



is measured in **centimetres**.

(3) Fill in the blanks using words centimetre, metre and kilometre.

(i) To measure the length of your science book in _____.

(ii) To measure the length of the chalk in _____.

(iii) To measure the length of your colour pencil in _____.

(iv) To measure the length of your classroom's window in _____.

(v) To measure the length of your classroom's door in _____.

Addition of units of length

Add measures of length in same units of length including abbreviations.

Using the examples given the teacher will show the students how to add similar units of length.

Let us learn how to add the same units of length with and without carrying.

Example : Add the following:

(1) $3 \text{ m} + 4 \text{ m}$

Solution:

$$\begin{array}{r} 3 \text{ m} \\ + 4 \text{ m} \\ \hline 7 \text{ m} \end{array}$$

Thus the sum is **7 m**.

(2) $36 \text{ km} + 15 \text{ km}$

Solution

$$\begin{array}{r} \textcircled{1} 36 \text{ km} \\ + 15 \text{ km} \\ \hline 51 \text{ km} \end{array}$$

Thus the sum is **51 km**.

EXERCISE 32

(A) Solve the following:

(1)

cm
67
+21

(2)

m
37
+85

(3)

km
28
+15

(4)

m	cm
11	33
+1	41

(5)

m	cm
8	68
+6	31

(6)

km	m
35	515
+12	436

(B) Add the following

(1) 66 cm and 23 cm

(2) 18 m and 9 m

(3) 23 km and 18 km

(4) 44 m 5 cm and 78 m 30 cm

Solve real life problems involving same units of length for addition with and without carrying.

Example 1:

Ali went to Rahim Yar Khan from Karachi. He travelled a distance of 756 km by train and 35 km by bus. How much total distance has he travelled?

Solution:

Distance travelled by train

Distance travelled by bus

$$\begin{array}{r} 756 \text{ km} \\ + 35 \text{ km} \\ \hline 791 \text{ km} \end{array}$$

Thus Ali travelled the distance **791 km**.

EXERCISE 33

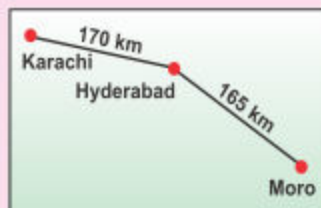
- (1) Wires of length 65 cm and 38 cm are needed to provide electricity in house. Find out the total length of wire required.



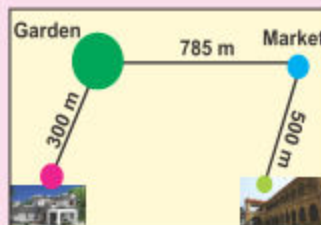
- (2) A shopkeeper bought 240 m of green cloth, 105 m blue cloth for his shop. What is the total length of the cloth bought?



- (3) Distance between Karachi and Hyderabad and Moro is shown in map. Find the distance from Karachi to Moro.



- (4) The distance covered by Ayaz cover daily from home to garden, garden to market and market to school, which is shown in the figure. Find the total distance covered by him.



Subtract measures of length in same units with and without borrowing Using the examples given the teacher will show the students how to subtract similar units of length.

Example:

Subtract 45 cm from 95 cm

Solution

$$\begin{array}{r} 95 \text{ cm} \\ - 45 \text{ cm} \\ \hline 50 \text{ cm} \end{array}$$

Thus the difference is 50 cm.



Activity

Subtract 786 m from 950 m.

Solution

$$\begin{array}{r} 950 \text{ m} \\ - 786 \text{ m} \\ \hline \text{m} \end{array}$$

Thus the difference is

EXERCISE 34

(A) Solve

(1)

$$\begin{array}{r} 81 \text{ cm} \\ - 42 \text{ cm} \\ \hline \end{array}$$

(2)

$$\begin{array}{r} 48 \text{ km} \\ - 19 \text{ km} \\ \hline \end{array}$$

(3)

$$\begin{array}{r} 26 \text{ m} \\ - 8 \text{ m} \\ \hline \end{array}$$

(4)

km	m
39	80
- 17	48

(5)

km	m
44	72
- 26	50

(6)

km	m
58	465
- 29	287

(B) Subtract the following:

(1) 36 cm from 76 cm

(2) 9 m from 12 m

(3) 88 km from 95 km

(4) 494 m from 680 m

(5) 864 km 150 m from 936 km 324 m

Solve real life problems involving same units of length for subtraction with and without borrowing



Activity

A shopkeeper sold **24 m** cloth from a roll of **33 m**. How much cloth is left in the roll?

Solution	Total cloth	33 m
	Cloth sold	– 24 m
	Cloth left	9 m

Thus the remaining cloth in the roll is m.

EXERCISE 35

- (1) An electrician used **35 m** wire for fitting of one fan in a class room of school from a roll of **50 m**. How much wire is left in the roll?



- (2) Shazia bought **72 cm** lace. She used **47 cm** lace to decorate a shirt. What length of lace is left with her?



- (3) A tailor master used **466 m** ribbon for making a pattern on a dress from **1100 m**. How much ribbon is left in the reel?



- (4) Height of Azhar is **1 m 22 cm** and height of his father is **2 m 50 cm**. How much is his father taller than him.



4.2 UNIT OF MASS/WEIGHT

Read standard units of mass/weight (kilogram and gram) including abbreviations. Weight is a force so should not be measured in kg/g

We have learnt that the weight (mass) of light object like a small paper clip, toffees, chillies and tea is measured in grams (g), the bulk quantity of objects like cement, wheat, flour etc is measured in kilogram (kg).

The standard unit of weight is kilogram (kg) which is equal to 1000gram (g).

$$1000g = 1kg$$

In daily life mostly shopkeepers use following weights



Activity

Colour the suitable unit of weight (kg, g) to measure the weight of the following objects:

Face Cream  ☐ g ☐ kg	Iron rod  ☐ g ☐ kg	Toothpaste  ☐ g ☐ kg
Ring  ☐ g ☐ kg	Biscuit Pack  ☐ g ☐ kg	Rice  ☐ g ☐ kg

Teacher's Note

The teacher will write the standard units of mass on the board along with their abbreviations. Teacher should revise the measure and write standard units of mass/weight and their abbreviation.

EXERCISE 36

- (1) Read the correct unit of mass (kg or g) for the following things. Colour the correct unit in each.



- (2) Fill in the blanks using kilogram or gram.

- (i) Weight of one apple is measured in g
- (ii) Weight of brick is measured in _____
- (iii) Weight of mobile phone is measured in _____
- (iv) Weight of log is measured in _____
- (v) Weight of necklace is measured in _____
- (vi) Weight of the hat is measured in _____

(3) Tick (✓) the correct unit of mass of the following things.

(i)



5 g	✓
5 kg	

(ii)



3 g
3 kg

(iii)



1 g
1 kg

(iv)



450 g
450 kg

(v)



6 g
6 kg

(vi)



500 g
500 kg

(vii)



300 g
300 kg

(viii)



38 g
38 kg

(ix)



25 g
25 kg

Add measures of mass/weight in same units with and without carrying.

Example: Add the following:

(1) $320\text{ g} + 210\text{ g}$

Solution:

$$\begin{array}{r} 320\text{g} \\ + 210\text{g} \\ \hline 530\text{g} \end{array}$$

(2) $3399\text{ kg} + 3577\text{ kg}$

Solution:

$$\begin{array}{r} \textcircled{1}\textcircled{1} \\ 3399\text{kg} \\ + 3577\text{kg} \\ \hline 6976\text{kg} \end{array}$$

EXERCISE 37

(A) Solve.

(1) $\begin{array}{r} 290\text{ g} \\ + 305\text{ g} \\ \hline \end{array}$

(2) $\begin{array}{r} 504\text{ g} \\ + 313\text{ g} \\ \hline \end{array}$

(3) $\begin{array}{r} 325\text{ g} \\ + 477\text{ g} \\ \hline \end{array}$

(4) $\begin{array}{r} 35\text{ kg} \\ + 46\text{ kg} \\ \hline \end{array}$

(5) $\begin{array}{r} 1214\text{ kg} \\ + 2453\text{ kg} \\ \hline \end{array}$

(6) $\begin{array}{r} 2396\text{ kg} \\ + 5725\text{ kg} \\ \hline \end{array}$

(B) Find the sum of following:

(1) $350\text{ g} + 200\text{ g}$

(2) $758\text{ kg} + 265\text{ kg}$

(3) $375\text{ g} + 900\text{ g}$

(4) $485\text{ g} + 7312\text{ g}$

Solve real life problems involving same units of mass/weight for addition with and without carrying.

Example :

Azhar has **678 g** of sugar in a jar. He puts another **245 g** more sugar in it. Find the total quantity of sugar in the jar.

Solution Sugar present in jar: $\overset{1}{6}\overset{1}{7}8 \text{ g}$

Sugar added: $+ 245 \text{ g}$

Total: 923 g



Hence the total quantity of sugar in the jar is **923 g**.

EXERCISE 38

- (1) Aslam bought **250 g** of black pepper and **350 g** red pepper for his kitchen. Find the total weight of both spices.



- (2) Calculate the weight of two bags of flour, when each bag weighs **10 kg**.



- (3) Kishwar buys **2 kg** of tomatoes, **7 kg** of onion and **6 kg** of cabbage. for a week. Find the total weight of all vegetables.



- (4) Imran purchased **28 kg** of cotton on Monday and **31 kg** of cotton on Tuesday. Find the total quantity of cotton he purchased.



Subtract measures of mass/weight in same units with and without borrowing.

Example 1: Subtract the following:

(i) $578\text{g} - 342\text{g}$

Solution

$$\begin{array}{r} 578\text{g} \\ - 342\text{g} \\ \hline \end{array}$$

$$\begin{array}{r} 578\text{g} \\ - 342\text{g} \\ \hline 236\text{g} \end{array}$$

(ii) $5302\text{kg} - 3415\text{kg}$

Solution

$$\begin{array}{r} 5302\text{kg} \\ - 3415\text{kg} \\ \hline \end{array}$$

$$\begin{array}{r} \overset{4}{5} \overset{12}{3} \overset{9}{0} \overset{12}{2} \text{kg} \\ - 3415 \text{ kg} \\ \hline 1887 \text{ kg} \end{array}$$

EXERCISE 39

(A) Subtract.

(1)

$$\begin{array}{r} 32 \text{ kg} \\ - 31 \text{ kg} \\ \hline \end{array}$$

(2)

$$\begin{array}{r} 496 \text{ g} \\ - 382 \text{ g} \\ \hline \end{array}$$

(3)

$$\begin{array}{r} 812 \text{ g} \\ - 653 \text{ g} \\ \hline \end{array}$$

(4)

$$\begin{array}{r} 724 \text{ kg} \\ - 345 \text{ kg} \\ \hline \end{array}$$

(5)

$$\begin{array}{r} 823 \text{ kg} \\ - 712 \text{ kg} \\ \hline \end{array}$$

(6)

$$\begin{array}{r} 986 \text{ g} \\ - 397 \text{ g} \\ \hline \end{array}$$

(B) Subtract.

(1) 348g from 600g

(2) 175kg from 428kg

(3) 125g from 345g

(4) 12kg from 20kg

(5) 150g from 370g

(6) 120kg from 200kg

Solve real life problems involving same units of mass/weight for subtraction with and without borrowing.

Example 1: Saleem bought **800 g** rice from market. His family consumed **450 g** rice. How much rice is left?

Solution:

Weight of rice purchased:	$\overset{7}{8}\overset{10}{0}0$ g
Weight of rice consumed:	$- 450$ g
Weight of rice left:	<u>350</u> g

Therefore weight of rice left is **350 g**.

EXERCISE 40

- (1) A grocer had **68 kg** of onion, he sold **53 kg** of onion in a day. How much onions are left with him?



- (2) A man bought **250 kg** of rice for a marriage ceremony. Out of which **150 kg** of rice was used. How much rice was left?



- (3) **5000 kg** of iron was purchased for the construction of a bungalow. Out of which **3550 kg** of iron was used. How much iron is left?



- (4) Farhan purchased **8 kg** mangoes. His brother Ikram purchased **30 kg** mangoes. Who purchased more and how much?



4.3 VOLUME/CAPACITY

We have learnt about capacity that capacity is the amount of liquid a container hold. We know that the standard unit of capacity is litre which is denoted by "l"

Starter: Using the examples given the teacher will discuss the units of volume and their abbreviations.

Units of volume

Read standard units of volume (litre and millilitre) including abbreviations

The standard units of capacity are litre (l) and millilitre (ml). Large quantity of liquid like water in water cooler or oil in a big bottle is measured in litres (l). Small quantity of liquids like glass of water or cup of tea, ink in pen is measured in millilitre (ml). There are 1000 millilitres in one litre.

1000 millilitres = 1 litre

or

1000 ml = 1 l



Activity

Colour the suitable unit of capacity for the given things of liquid.



ml l



ml l



ml l



ml l



ml l



ml l



ml l



ml l

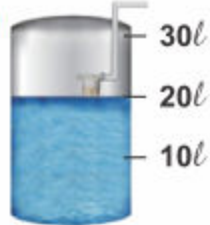
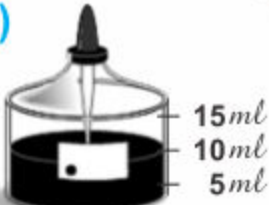
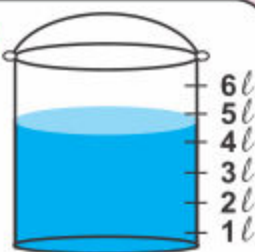
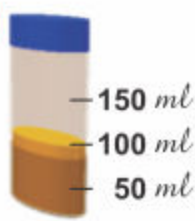
Teacher's Note

Teacher should revise the standard units of volume and their abbreviation.

Measure and write standard units of volume including abbreviations

**Activity**

Read and write the volume of liquids that the following objects contain.

(1)  20 litre	(2)  _____ millilitre	(3)  _____ litre
(4)  _____ litre	(5)  _____ millilitre	(6)  _____ millilitre
(7)  _____ litre	(8)  _____ millilitre	(9)  _____ millilitre

Teacher's Note

Teacher should help the students to measure the units of volume.

EXERCISE 41

(A) Which unit would be used to measure?

①



90 ml
900 ml
2900 ml

②



3 l
30 l
300 l

③



50 ml
500 ml
5000 ml

(B) Write the appropriate amount of liquid.

①

Rizwana filled a bucket with water to clean her room. Tell that the bucket hold 9 litre or 9 millitre of water.

9l



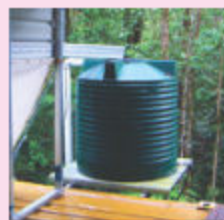
②

Aabish adds one tea spoon of vanilla to her cake recipe. Did she use 5ml or 5l of vanilla?



③

Your school has a large tank of water. The tank will hold 100 l or 100 ml of water?



④

Farzana bought juice for her friends to drink at her birthday party. The volume of juice is 250 l or 250 ml?



⑤

Ali bought a cup of tea. The cup hold 200 l or 200 ml of tea?

